

Day 66 – DHCP Configuration & Automated IP Addressing

Overview

Day 66 introduced **DHCP (Dynamic Host Configuration Protocol)**, a network service that automatically assigns IP addresses and other network settings to client devices.

In previous labs, every PC was configured manually with:

- IP address
- Subnet mask
- Default gateway

In this lab, the routers were configured as DHCP servers, allowing the PCs to automatically receive their network settings when switched to DHCP mode.

This lab demonstrated how modern enterprise and cloud networks automate device configuration and eliminate the need for manual IP assignments.

Objectives

By the end of this lab, I was able to:

- Understand the purpose of DHCP
 - Explain the DHCP DORA process
 - Configure DHCP pools on Cisco routers
 - Exclude reserved addresses from DHCP assignment
 - Automatically assign IP addresses to clients
 - Verify DHCP leases
 - Test connectivity across multiple routed networks
 - Integrate DHCP with an existing RIP-based network
-

DHCP Concepts

What is DHCP?

DHCP automatically provides:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

without requiring manual configuration on every device.

DORA Process

DHCP uses a four-step process known as **DORA**:

Discover

Client → "Is there a DHCP server available?"

Offer

Server → "I can offer you an IP address."

Request

Client → "I would like to use that address."

Acknowledge

Server → "Approved. The address is yours."

Existing Network Topology

This lab used the completed Day 65 topology.

LAN10 LAN20 LAN30

PC0 PC1 PC2

|||

SW0 SW1 SW2

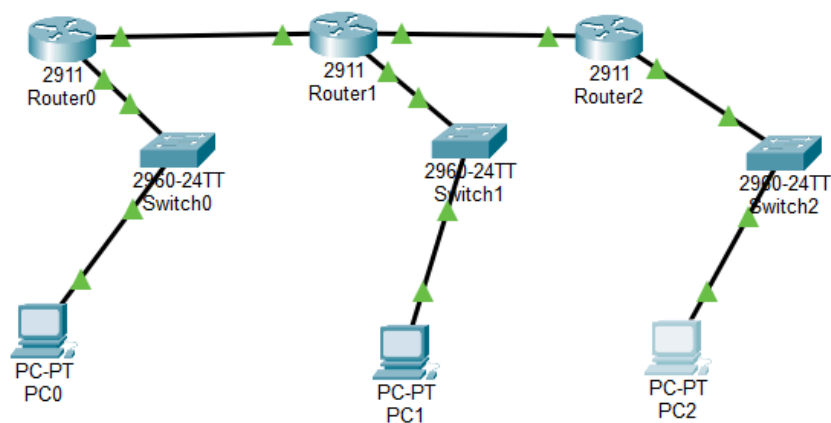
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R0 ----- R1 ----- R2

RIP Version 2 remained active and provided routing between all LANs.

Screenshot #1 – Complete Day 66 Topology

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Highlight

- Three routers
 - Three switches
 - Three PCs
 - Router-to-router links
 - RIP-enabled network
 - Green connectivity indicators
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Network Addressing Plan

LAN	Network	Gateway
LAN10	192.168.10.0/26	192.168.10.1
LAN20	192.168.20.0/26	192.168.20.1
LAN30	192.168.30.0/26	192.168.30.1

DHCP Configuration

Router0 DHCP Pool

```
ip dhcp excluded-address 192.168.10.1
```

```
ip dhcp pool LAN10
```

```
network 192.168.10.0 255.255.255.192
```

```
default-router 192.168.10.1
```

```
dns-server 8.8.8.8
```

Purpose

- Reserves Router0's gateway address
 - Assigns addresses in LAN10
 - Provides gateway and DNS information
-

Router1 DHCP Pool

```
ip dhcp excluded-address 192.168.20.1
```

```
ip dhcp pool LAN20
```

```
network 192.168.20.0 255.255.255.192
```

```
default-router 192.168.20.1
```

dns-server 8.8.8.8

Router2 DHCP Pool

ip dhcp excluded-address 192.168.30.1

ip dhcp pool LAN30

network 192.168.30.0 255.255.255.192

default-router 192.168.30.1

dns-server 8.8.8.8

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PC Configuration

Each PC was changed from static addressing to DHCP.

Navigation

Desktop

→ IP Configuration

→ DHCP

The PCs automatically requested and received addresses from their local router.

Screenshot #2 – PC0 DHCP Configuration

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☒ DHCP	☐ Static
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1
DNS Server	8.8.8.8

Highlight

IP Address: 192.168.10.2

Gateway: 192.168.10.1

DNS Server: 8.8.8.8

Screenshot #3 – PC1 DHCP Configuration

☒ DHCP	☐ Static
IPv4 Address	192.168.20.2
Subnet Mask	255.255.255.192
Default Gateway	192.168.20.1
DNS Server	8.8.8.8

Highlight

IP Address: 192.168.20.2

Gateway: 192.168.20.1

DNS Server: 8.8.8.8

Screenshot #4 – PC2 DHCP Configuration

☒ DHCP	☐ Static
IPv4 Address	192.168.30.2
Subnet Mask	255.255.255.192
Default Gateway	192.168.30.1
DNS Server	8.8.8.8

Highlight

IP Address: 192.168.30.2

Gateway: 192.168.30.1

DNS Server: 8.8.8.8

DHCP Lease Verification

The DHCP leases were verified using:

```
show ip dhcp binding
```

Router0 Lease

192.168.10.2

Router1 Lease

192.168.20.2

Router2 Lease

192.168.30.2

After DHCP configuration, inter-LAN communication was tested.

PC0

ping 192.168.20.2

ping 192.168.30.2

Successful.

PC1

ping 192.168.10.2

ping 192.168.30.2

Successful.

PC2

ping 192.168.10.2

ping 192.168.20.2

Successful.

Screenshot #8 – PC0 Remote Ping Test

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```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time=1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Highlight

- Successful reply
 - DHCP-assigned IP address
 - Routed communication
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Screenshot #9 – PC2 Remote Ping Test

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```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=125
Reply from 192.168.10.2: bytes=32 time<1ms TTL=125
Reply from 192.168.10.2: bytes=32 time=2ms TTL=125
Reply from 192.168.10.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Highlight

- Communication across multiple routers
- RIP and DHCP functioning together

Relationship Between Day 65 and Day 66

Day 65

Dynamic Routing (RIP)

Routers learned routes automatically.

Day 66

DHCP

Clients learned addresses automatically.

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Together, these created a much more realistic enterprise network environment.

What I Learned

- DHCP automates IP address assignment.
 - DHCP reduces manual configuration effort.
 - The DORA process governs DHCP communication.
 - Routers can function as DHCP servers.
 - Excluded addresses prevent conflicts with gateway addresses.
 - DHCP works alongside RIP routing.
 - Network services build on top of a functioning routed network.
 - Automated addressing is used in enterprise and cloud environments.
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Troubleshooting Notes

Some initial ping tests showed 25% packet loss.

This occurred because the first packet was used for ARP resolution while devices learned MAC address information.

Subsequent packets succeeded, confirming that routing and DHCP were functioning properly.

This reinforced the importance of understanding the difference between temporary ARP behavior and actual network failures.

Reflection

Day 66 introduced my first true network service. Instead of manually assigning IP addresses to every device, I configured routers to automatically provide addressing information through DHCP. After verifying DHCP bindings and testing communication between all three LANs, I confirmed that the service was functioning correctly alongside RIP dynamic routing. This lab demonstrated how enterprise networks simplify device deployment and prepared me for future topics such as DNS, NAT, ACLs, and integrated network services.